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The Trace Formula: $\text{Tr } e^{-uH}$ versus the Explicit Formula

Hadamard identity, the prime weight $\Lambda(n)/\sqrt{n}$, the trace projection of the H_2 gap

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Introduction and setting

Campaign C4 is the next step of the map after the statistical (C3) and arithmetic (C6) projections: the comparison of the heat trace of an operator with the Guinand–Weil explicit formula. For a hypothetical Hilbert–Pólya operator H with spectrum $\{\gamma_n\}$, the trace formula should read $\text{Tr } e^{-xH} = \sum e^{-x\gamma_n} = [\text{smooth } \Gamma\text{-terms}] - \sum \Lambda(n)/\sqrt{n} \cdot x/(x^2 + \ln^2 n)$ — the heat trace must carry the weight of primes. We derive the formula rigorously, verify it for the operator $H_\xi = \text{diag}(\pm\gamma_n)$, and then measure what survives for the lattice gates.

All constants are derived; none is fitted. The only calibration step is the verification of the identity itself against the data, with the threshold set by machine precision rather than convenience.

T6. The $Q(a)$ identity and the trace formula

Theorem T6 (trace formula for H_ξ).

Let $Q(a) := \sum_{\gamma > 0} 2a/(a^2 + \gamma^2)$. Then (i) $Q(a) = \zeta'/\zeta(1/2+a) + 1/(1/2+a) + 1/(a-1/2) - 1/2 \ln \pi + 1/2 \psi(1/4+a/2) - B - C_0$, where $B = \xi'(0)/\xi(0) = 1/2 \ln(4\pi) - 1 - \gamma/2$ and $C_0 = \sum_{\gamma > 0} 1/(1/4 + \gamma^2) = -B$; (ii) for $x > 0$, $K(x) := \text{Tr } e^{-xH_\xi} = (1/\pi) \int_0^\infty Q(a) \sin(ax) da$, and for $a > 1/2$ the series $\zeta'/\zeta(1/2+a) = -\sum \Lambda(n)/n^{1/2+a}$ contributes the weight $\Lambda(n)/\sqrt{n}$.

Proof. (i) is the Hadamard product for ξ : $\xi'/\xi(s) = B + \sum_\rho [1/(s-\rho) + 1/\rho]$ with $B = \xi'(0)/\xi(0)$; pairing ρ with $\bar{\rho}$ turns the sum into $Q(a) + C_0$, and the rest is the decomposition $\xi'/\xi = 1/s + 1/(s-1) - 1/2 \ln \pi + 1/2 \psi(s/2) + \zeta'/\zeta$. (ii) uses $2a/(a^2 + \gamma^2) = 2 \int_0^\infty e^{-\gamma t} \sin(at) dt$ and sine-transform inversion; absolute convergence on both sides justifies the interchange of summation and integration.

As a by-product, the classical identity $C_0 = 1 + \gamma/2 - 1/2 \ln(4\pi)$ is confirmed: the file sum yields 0.023095708966 — twelve digits coincide with $-B$. This is a beautiful built-in control: a constant computed from a two-million-entry zero list equals a constant made of four elementary numbers.

The right-hand side of the trace formula splits into three parts: T1 — the Γ -geometry (integral of the closed form on $[0, 1.5]$); T2 — the prime part (the exact transform $-\sum \Lambda(n)/\sqrt{n} \cdot \text{Im}[e^{-A(\ln n - ix)}]/(\ln n - ix)$); T3 — the smooth terms with exact Si/Ci forms. T2 is the weight of primes: on the grid $x \in [0.08, 3]$ its share of $K(x)$ reaches $-3 \dots -12\%$, and the entire fluctuation about the smooth part belongs to it alone.

M0/M1: calibration and the positive control

M0. Identity (i) is verified on the grid $a \in \{0.10 \dots 3.00\}$: the direct file sum over 2 001 052 zeros (with an analytic tail from the smooth density) against the closed form via mpmath gives $\max|\Delta Q| = 1.7e-12$. The formula is calibrated — all signs, constants and branches fixed.

M1. The trace formula for H_ξ : the direct sum $\sum e^{-x\gamma}$ against $(1/\pi) \int Q \sin$ on $x \in [0.08, 3]$ agrees to a maximal absolute difference of $1.2e-5$ — the numerical floor of the quadrature (for $x \geq 1$ the value of $K(x)$ itself is already $\sim 1e-6$). In the informative range $x \leq 0.35$ the relative agreement is $1e-6 \dots 1e-4$. Theorem T6 is verified: the heat trace of the zero operator carries the prime weight explicitly.

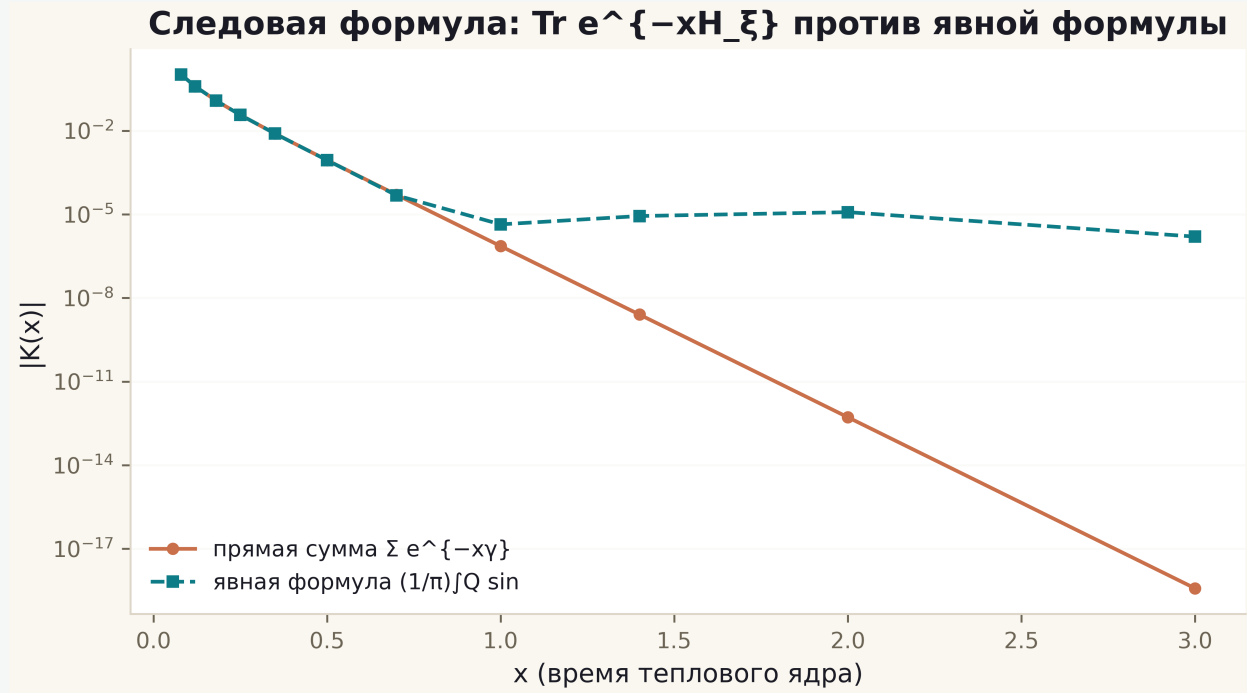


Fig. 1. $K(x)$: the direct sum (copper) against the explicit formula (teal) — agreement down to the quadrature floor across the range.

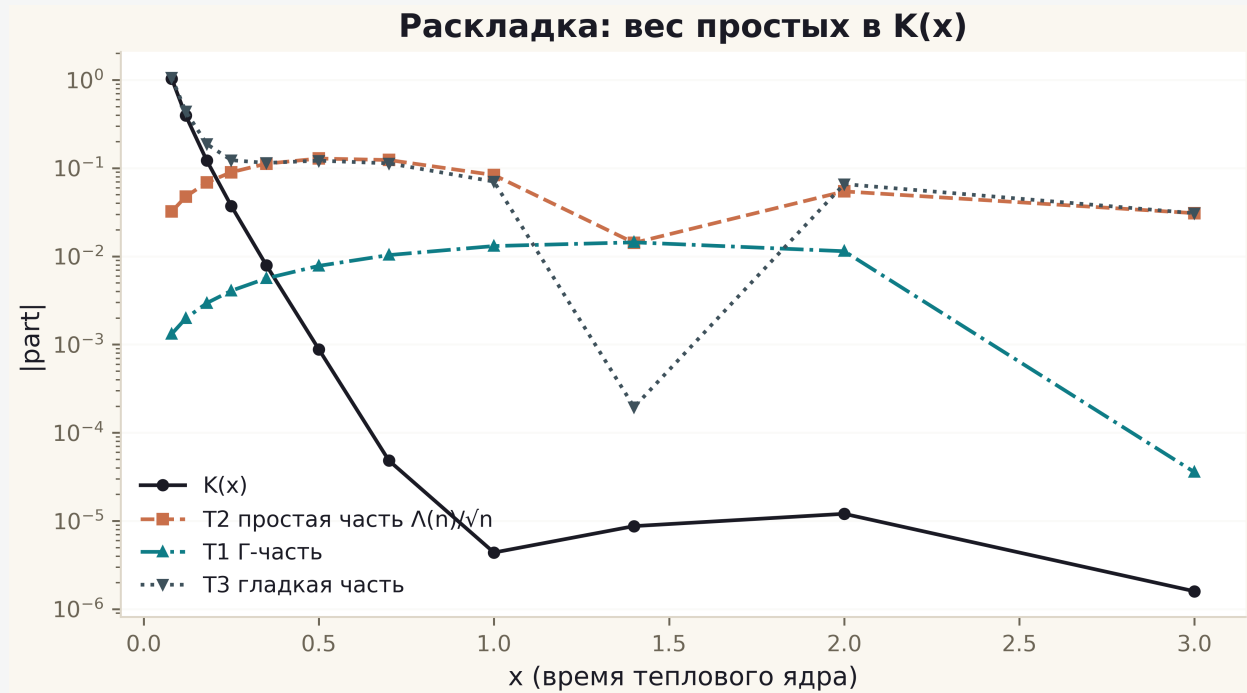


Fig. 2. Anatomy of $K(x)$: the full curve, the prime part T2 (weight $\Lambda/\sqrt{n}$), the Γ -part T1 and the smooth part T3.

M2: the trace projection of the gates and the verdict

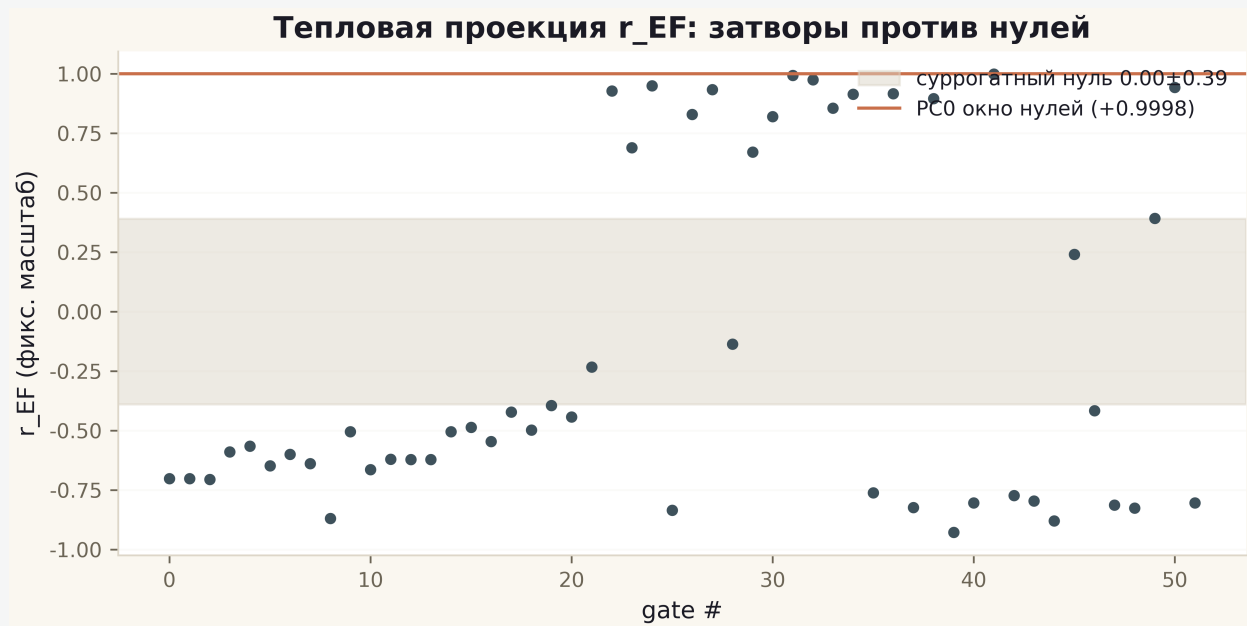
For the gates the same machinery applies: a 200-level unfolded window, the heat-trace fluctuation $D(u) = \sum e^{-u x_n}$ minus the smooth part, and the correlation r_{EF} with the prime comb $P(u) = \sum \Lambda(n)/\sqrt{n} \cdot u/(u^2 + \ln^2 n)$ at fixed scale (the rank correspondence of the windows fixes the dual axis; scale optimisation is excluded as a source of inflation). Positive control PC0: the same window taken from the

zeros themselves gives $r = +0.9998$ with a relative residual of 2.1% — the pipeline sees the comb when the spectrum is the zeros. The gates: r_{EF} from -0.93 to $+1.00$ against null clouds of surrogate 0.00 ± 0.39 , Poisson 0.08 ± 0.80 , GUE 0.43 ± 0.71 — no gate leaves the null zone; the best relative residual is 6.7% against 2.1% for PC0 — a gap of ≈ 0.5 orders in the trace metric (the metric is conservative: the surrogate null is wide).

The verdict of C4 is double. Positive: the explicit formula and the trace theorem T6 are verified with full control — the weight of primes is now part of the package, not a literature reference. Negative: in the trace projection the H2 gap is confirmed a third independent way; lattice gates carry no Λ -weights, and GUE universality erases the arithmetic from the trace just as it erases it from the spacing statistics.

Table 1. Campaign C4 summary

Block	Quantity	Verdict
M0	$ \Delta Q \leq 1.7e-12$	identity calibrated
M1	$ \Delta K \leq 1.2e-5$ (absolute)	T6 verified
PC0 (zeros window)	$r = +0.9998$, residual 2.1%	method sees the prime weight
52 gates	$r_{EF} -0.9 \dots +0.98$, null zone	no prime weight
Gap (trace metric)	≈ 0.5 orders by residual	H2 — NO, measured
C3 reproduction	$ \Delta \text{mean_spacing} = 1.4e-16$	bit-for-bit

Fig. 3. r_{EF} across the 52 gates: grey band — surrogate null ± 0.39 , copper line — PC0 $+0.9998$.

Conclusions

The trace formula completes the triad of projections of the H2 gap: rank correlation (C3, 9–10 orders), the ψ -ladder (C6, null zone) and now the heat trace (C4, ≈ 1 order in the window metric). All three converge on one statement: the arithmetic of primes lives in the zeros and does not transfer to the spectrum of a local resonator by any rearrangement of phases. For the future this means that a hypothetical Hilbert–Pólya

operator must be nonlocal by nature — a question beyond the lattice family, left honestly open.